Split plan: moving the prognostic material and the F-family out of paper 02

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Status. Plan only. No manuscript, driver or data file has been changed. Line numbers refer to `report.Rmd` at commit 8e810ca9d (2026-10-29). They will drift as soon as the first cut is made, so work from the section labels.

1. Decision

Paper 02 is narrowed to one question: how to represent carryover and choose the analysis half-life for the biomarker-treatment interaction when the biomarker is predictive only. The data come from the refined process under covariance moderation, meaning the biomarker-response correlation is the interaction mechanism.

Everything about a biomarker that is also prognostic moves to a new companion paper, provisionally paper 15. So does the F-family (F1, F1A, F1A0, F1A0B), whose expectancy terms exist to protect against a prognostic biomarker.

A note on wording: the narrowed paper 02 does not use the published process at 58b32a9. Step coupling in that process removes the signal at any positive half-life (`docs/55`). Describe the scope as "the refined process under covariance moderation, with a predictive-only biomarker".

Rationale.

- The split removes two of paper 02's submission blockers outright:
  - the repaired strength-0.6 rows
  - the untested bm:t remedy

  Both become properly scoped results in paper 15, where the 0.6 cells are the repair-artifact exhibit and the bm:t check becomes a tested analysis.
- It cuts about 15 to 20% of the manuscript.
- It gives the prognostic material a coherent paper of its own. That paper is largely drafted already across `docs/47`, `docs/49`, `docs/56`, `docs/64`, `docs/65` and `docs/series/ch19`.

1a. Decisions needed before work starts

Three points emerged while mapping the split. Each needs an author decision; the detail is in the section cited.

1. **Which F-family member carries the joint recommendation (Section 4).** As defined, F1 fixes the analysis half-life at 0.2 week. In paper 02's cells the true half-life is 0.5 to 2 weeks, so F1 is misspecified there and loses power. "Use F1" would therefore contradict paper 02's own evidence. The candidates are:

   - **F1A0B:** half-life by AIC on the null model, bootstrap calibrated. This is the current provisional recommendation.
   - **F1A:** half-life by AIC on the interaction model, which is G9's rule. CLAUDE.md calls this pattern liberal.
   - **F1 at a fixed half-life from prior pharmacology**, rather than the 0.2-week policy value.

   The choice also changes the CLAUDE.md "Recommended analysis" paragraph.
2. **Large-N size evidence for G9 (Section 2.6).** Family 6, the $N \in \{140, 280\}$ cells, is entirely prog-PB. After the split, paper 02 has no G9 size evidence above $N = 70$. There are two options:

   - **(a)** State the $N \leq 70$ limit, which the paper already partly does.
   - **(b)** Run a confound-free large-N null. The cost is a rough estimate of 1 to 3 hours on zeus, not a timing.
3. **Overlap between paper 01 Study 3 and paper 15 (Section 5.2).** Paper 01 already contains a prognostic stress test and a joint feasibility table (`tab:prognostic`). The recommendation is that paper 15 cite paper 01 for the DGP-level facts and present only the analysis-level consequences. Paper 01, which is near submission at grade B, would not be reopened.

2. Paper 02 after the split

2.1 Scope statement (Introduction and Methods)

Add one paragraph to the Introduction. Suggested text:

> The biomarker in this study is predictive only: it changes the drug response and nothing else. A biomarker that is also prognostic raises a different question, namely whether the interaction test confuses an association with the natural history or the placebo response with moderation of the drug effect. A constant prognostic shift is absorbed exactly by the biomarker main effect of every analysis here and cannot move the interaction estimate. Prognostic associations that vary over time, and the design-expectancy terms that protect against them, are the subject of a companion paper [paper 15].

Keep the absorption check for the constant shift (b_{tv}, "prog-TV-grow" in the stored data). It takes one sentence and answers the first question a referee will ask.

2.2 Section-by-section cut list

Status values:

- **Delete:** remove the section.
- **Trim:** remove the prognostic or F1 content only.
- **Rewrite:** recast the text without it.
- **Keep:** leave as it is.
- **Move:** relocate the material to paper 15.

Section (label)	Lines	Action
Setup chunk: `s28`, `s30`, `pdc`, `ceil` loads	122-129	Delete. These are the prognostic stress, random-effects, PD-check and ceiling data.
Setup chunk: `m_lab`, `focus`, `focus_cols`, `rv_lab`	140-164	Trim. Remove F1, F1A, F1A0 and F1A0B.
Setup chunk: `rvT`	168	Delete. Keep `rvS`; see the slope row below.
Abstract	230-273	Rewrite: (a) Methods: drop "or add expectancy terms (F1, F1A0B)". (b) Results: drop the F1 sentence, the trend-bias sentence and the strength-0.6 artifact sentence. (c) "Eight scenario families" becomes six; see 2.3. (d) Conclusions: add "for a biomarker that is predictive only".

Section (label)	Lines	Action
Introduction, ladder overview	385-400	Trim. Remove the F-family sentence and the prognostic, trend and large-N items from the list of stresses. Add the scope paragraph from 2.1.
Aims (`sec-aims`)	457-480	Trim. Remove "prognostic pathways, large samples, a biomarker that predicts the natural-history trend" from the second aim. Keep "patient-specific slope variance".
Table `tab-families`	482-500	Rewrite. Delete rows 5 (prognostic) and 6 (large N; every cell there is prog-PB). Row 7 becomes slope variance only. Renumber.
`sec-ladder-def`	643, 666	Trim. Remove the pointers to the F-family.
`sec-e0`	673-688	Keep. E0 contains D_e because Hendrickson's analysis does. That is part of the original analysis, not an F-family feature. Add one clause noting D_e is constant, and so dropped, under CO.
`sec-comparators`	803-839	Rewrite. Delete the F-family paragraph (807-831). Keep G9-null and G8 at a fixed half-life. Retitle "Side analyses: variants of the selection".

Section (label)	Lines	Action
`sec-pathways`	841-866	Delete, except one sentence on constant-shift absorption, which goes in the scope paragraph.
`sec-choice`	868-886	Keep. The earlier bm:L and bm:tsd models are a carryover question.
`sec-grid`	924-960	Trim. Remove the prognostic-cell clauses (924, 929, 950-955).
`sec-corrections`	1008-1058	Trim. The PD-refusal correction stays as a statement about the driver. Its prognostic-scenario evidence moves to 15 (1041-1044).
Ladder-at-a-glance (`sec-glance`) and Fig. 1-3 captions	1072-1274	Rewrite after the ladder script is rerun (2.4). Remove F1 and F1A0B from the captions (1127, 1159) and from the prose (1206-1207, 1271).
`sec-rung1`, E0 isolation	1282-1580	Keep. The expectancy mentions here describe E0's D_e term, which stays.
Rung 4 null variants (`stage3-variants-null`)	2077-2125	Trim. Remove the F1 column.
`sec-alternatives`	2172-2341	Rewrite. Keep the fixed-half-life and null-model-selection alternatives. Delete the F-family evidence (2176-2183, 2279-2339, the 'G9>F1' paired step). Move the G9-versus-F1A0B comparison to 15.
`sec-scen-grids`	2353-2457	Trim. Remove the F1 columns (2381-2394, 2436-2448).

Section (label)	Lines	Action
`sec-dropout`	2458-2585	Trim. Remove the F1 columns (2472-2483, 2499-2500, 2514, 2524-2531, 2565-2566). Line 2533 cites the no-dropout reference of the prognostic stress job. Re-anchor it on a non-prognostic reference or drop the comparison.
`sec-prognostic` (all subsections, including `sec-feasibility`, `sec-feasible-grid` and "Larger samples")	2586-2939	Move to 15.
`sec-trend`: trend pathway	2940-3012	Move to 15 (table `stage5-trend`, Fig. 4).
`sec-trend`: patient slope variance	3014-3045	Keep as its own short subsection, "Patient slope variance", with G8 and G9 only. It is a heterogeneity stress, not a prognostic one: the slope is unrelated to the biomarker. Delete the fig4 trend contrast (3028-3040).
`sec-random`	3046-3090	Move to 15. It was fitted on the prognostic stress cells, 8 of 12 of them the repaired strength-0.6 cells, and was motivated by the artifact.
`sec-calibration`	3091-3178	Trim. Remove the F1 kappa values (3142-3143).

Section (label)	Lines	Action
`sec-which`, ladder score narrative	3179-3262	Rewrite from the regenerated table B. "G9 has the largest cell size … under the trend pathway" no longer applies. Remove the F1 power-loss sentence (3250).
Discussion, `sec-bought`	3265-3364	Trim. Remove the F-family and prognostic summaries (3331-3359).
`sec-recommend`	3365-3414	Rewrite. State the scope condition (predictive-only biomarker). Insert the joint-recommendation sentence (Section 4). Remove the trend caveat (3404-3406); it moves to 15.
"What the expectancy terms are for"	3415-3442	Move to 15. In 02, replace it with the one-sentence expectancy note in 2.5.
Limitations	3443-3497	Trim. Remove the repaired-cell, prognostic-grid and large-N F-family sentences (3469-3488). Add the large-N gap (2.6).
Conclusions	3534-3595	Rewrite items 5, 6 and 8. Item 5 keeps the fixed-half-life and null-selection findings only. Item 6 (feasibility) moves to 15. Item 8 loses the prog-PB and trend sentences and gains the scope condition.

Section (label)	Lines	Action
Glossary	3596-3715	Trim. Remove F1, F1A, F1A0, F1A0B, the prognostic labels and the feasible-grid note (3598, 3625, 3642).
Reproducibility	3751-3878	Trim. Remove the jobs 28 and 30, scripts 30 and 31, and block T entries (3786-3792, 3829).

2.3 Scenario families after the split

New	Old	Content
1	1	Original 30-cell grid
2	2	Misspecified half-life and decay shape
3	3	Serial correlation
4	4	Patient-level dropout
5	7 (block S only)	Patient slope variance
S	S (block N: $c_{pb} = 0$ cells only; block A)	Independent-seed Hybrid run

Block N of the referee variants holds four null cells at $c_{pb} = 0.3$. They leave family S. The pooled G9 null figure in the abstract ("7,987 null fits … 0.055") must be recomputed without them. Check whether `m2-pooled-size.csv` (script 34) already pools only $c_{pb} = 0$; the ladder script sets `pooled_null = c_pb == 0` for block N, but script 34 has not been checked.

2.4 Ladder script (`37-ladder-tables-figures.R`)

The script reads stored data only, so regenerating it is cheap. The changes are:

1. Drop family 5 (job 28 and the gap-fill sub `C1-ladder`), family 6 (sub `C2-largeN`), block T, and the block N cells with $c_{pb} > 0$.
2. Drop F1 and F1A0B from `analysis_cols`, the paired steps (the 'G9>F1' row) and `side_x`. Drop them from the Fig. 2 and Fig. 3 analysis lists.
3. Delete Fig. 4 (trend). Paper 15 will need its own version.
4. Rerun the script and check that tables A, B and C and Fig. 1-3 still satisfy the Rmd's `stopifnot` guards.

5. Re-read every inline `lv()`, `inc()` and `bv()` call against the new tables.

Rather than deleting code, add a `--scope predictive` switch that filters these out. Paper 15 can then run the same script with `--scope all` for its G-versus-F comparisons.

2.5 Expectancy note for 02

A referee will ask why the G-family omits a design covariate that varies in Hybrid and OL+BDC. Answer in one sentence in `sec-recommend`:

> With a predictive-only biomarker the G-family holds nominal size without the design-expectancy terms in every family examined; the terms matter when the biomarker may also be prognostic, which the companion paper [15] examines.

Do not claim more than that. The claim that F1 matched G9's power at every alternative belongs to paper 15, which carries the evidence.

2.6 Evidence gap the split opens

Family 6 (large N) consists entirely of prog-PB cells. After the split, paper 02 has no G9 size evidence above $N = 70$ in the predictive-only case. Choose one of two options:

- **(a) State the limit.** Size is verified to $N = 70$. This matches what the paper already says ("measured only up to $N = 70$ with complete data").
- **(b) Run a confound-free large-N null.** Hybrid, $N \in \{140, 280\}$, $c_{bm} = 0$, $c_{pb} = 0$, G8 and G9, 1,000 replicates, through `32-run-g9-gap-fill.R` or a new block of `36-run-referee-variants.R`. The section the cut removes already asked for this check ("a targeted size check at larger N with a confound-free counterpart"). Cost is not timed. G9 refits five models per replicate, and at $N = 280$ each fit is several times slower than at $N = 70$. A rough estimate is 1 to 3 hours on zeus. Time it with `--time` first.

Option (a) is sufficient for submission. Option (b) is cheap and makes the paper stronger.

2.7 Effort for 02

- 1 to 2 days of rewriting.

- About 1 hour for the ladder script and checks.
- No new simulation unless 2.6(b) is chosen.

Expected grade after the cut: B, with the front matter (cover letter, title page, declarations) still outstanding as in `docs/66`.

3. Paper 15 outline: prognostic biomarkers and the interaction test

Working title: *When the biomarker is also prognostic: bias in the biomarker-treatment interaction test of aggregated N-of-1 trials and two remedies.* Directory: `analysis/report/15-prognostic-biomarker/`. Target: the same venue as 01 and 02, as a third companion, or a methods journal on its own.

Section	Content	Source (existing)
1. Introduction	Predictive versus prognostic biomarkers; why the interaction test can confuse the two in N-of-1 designs with varying expectancy	`docs/49`, `ch19`, lexicon `docs/29`
2. Taxonomy of pathways	Constant shift (absorbed); correlation pathways prog-BL, prog-TV-const and prog-PB; mean-shift trend pathway (unbounded)	02 `sec-pathways`; `tidyverse/R/functions.R` trend block
3. Feasibility	Closed-form joint bound on the prognostic correlation and c_{bm} (Schur complement); feasible grid	`docs/65`; scripts 30 and 31; 02 `sec-feasibility`. **Paper 01 already has a joint-ceiling table** (`tab:prognostic`, `18-prognostic-joint-ceiling.R`). Cite 01 for it rather than duplicating; see 5.2.
4. The repair artifact	Strength-0.6 cells: power loss from silent PD repair, not confounding	`docs/65`; jobs 28 and 30; 02 `sec-prognostic` "strength-0.6" subsection; `sec-random`

Section	Content	Source (existing)
5. Analyses	G-family (G8, G9) against the F-family (F1, F1A, F1A0, F1A0B); expectancy terms D_e, $bm{:}D_e$; why they absorb the expectancy-scaled on/off contrast under prog-PB	`docs/56`, `docs/57`, `docs/62`, `docs/64`; 02 `sec-comparators` and "What the expectancy terms are for"
6. Correlation pathways: results	Size and power at strength 0.3 and on the feasible grid; large N under prog-PB	Job 28; gap-fill `C1-ladder` and `C2-largeN`; 02 tables `stage5-*`
7. Trend pathway: results and the bm:t remedy	Bias of every analysis; linear and quadratic bm:t; cost when there is no trend	Block T (job 36); **new run** (3.1)
8. G-family against F-family without a prognostic effect	The F-family's cost when unneeded; F1's fixed-half-life penalty versus F1A0B	Job 27 (`tier1-g-vs-f-e0`), job 21 (`g-vs-f-comparison`); 02 `sec-alternatives` F-material
9. Recommendation	Joint recommendation (Section 4)	
Appendix	Constant-shift absorption proof; repair mechanics	`docs/65`

3.1 New simulation for paper 15

bm:t block.

- Add specifications `E2cr2_bmt`, `E7cr2_bmt` and `F1_bmt` to `simulation-core.R`: the existing formula plus `bm:t`. For G9, add the term to both the AIC-selection formula and the refit formula. Optionally add quadratic variants.
- Add a new block to `36-run-referee-variants.R`. Cross the trend pathway $b_{tv,trend} \in \{0, 0.3\}$ with $c_{bm} \in \{0, 0.45\}$ at Hybrid, $N = 70$, $t_{1/2} = 0.5$, 1,000 replicates, plus a 5,000-replicate null trend cell for size.
- Estimate about 30 to 45 minutes on zeus. Block T took 8 minutes for 2 cells and 3 specifications.

A local pilot on 2026-10-29 (scratch script, 300 replicates, G8 only) gave the following:

- The null bias of +0.26 fell to +0.10 with linear bm:t and to +0.00 with quadratic.
- Power fell from 0.96 to 0.94 and 0.92.

The DGP's trend follows the TV Gompertz course, not a straight line, so linear bm:t is expected to be only a partial remedy.

3.2 Effort for paper 15

About 1 to 2 weeks:

- Drafting is mostly assembly from the sources above.
- One new simulation block (about 1 hour of compute plus half a day of code).
- The ladder script runs with `--scope all` for the G-versus-F figures.

4. Joint recommendation (both papers, CLAUDE.md, NOTATION.md)

The two papers must give one answer between them. Present the F-family as the G-family plus D_e and $bm{:}D_e$:

- **Paper 02** settles the carryover layer: an exposure-weighted predictor, the half-life chosen by AIC or fixed, and CR2 with Satterthwaite df.
- **Paper 15** adds the expectancy layer when the biomarker may be prognostic, and a bm:t layer if the biomarker may predict the natural-history trend.

Open decision: which F-family member carries the joint recommendation. F1 at a fixed 0.2-week half-life is CLAUDE.md's validated fallback. In paper 02's cells, however, it loses power whenever the true half-life is 0.5 to 2 weeks, because 0.2 is then misspecified. "Use F1" would therefore contradict 02's evidence. There are three candidates:

1. **F1A0B** (half-life by AIC on the null model, bootstrap calibrated). This is CLAUDE.md's provisional recommendation and the closest analog of G9-null with expectancy terms. `docs/64` found no Type I inflation and no power gain over F1.

2. **An F-family member with G9's rule**: AIC on the interaction model (F1A), with CR2. This is the closest analog of G9, but CLAUDE.md calls the F1A pattern liberal. Paper 02 measures that cost for G9 at about half a point of size.
3. **F1 at a fixed half-life chosen from prior pharmacology** rather than the 0.2-week policy value. Paper 02 supports a fixed 1-week value for the G-family.

Recommended wording, pending that decision:

> For a biomarker believed to be predictive only, use G9 (paper 02). Where the biomarker may also be prognostic, add the design-expectancy terms D_e and $bm{:}D_e$ (paper 15), which cost little power when they are not needed; where it may predict the rate of natural-history change, also add a biomarker-by-time term.

Once decided, update:

- the CLAUDE.md "Recommended analysis for the interaction (F1)" paragraph
- NOTATION.md (analysis IDs; the new paper number)
- `docs/66` (portfolio table)
- `docs/60` Part 2 (paper 02 plan)

5. Cross-paper effects

5.1 Paper 01

Paper 01 cites paper 02. Check that none of its citations points at material that moves. Grep for "companion" in 01. As of 8e810ca9d, 01's own prognostic content is its Study 3 stress test, which is independent of 02.

5.2 Overlap between paper 01 Study 3 and paper 15

Paper 01 already contains:

- a stress test under a prognostic biomarker (Study 3, `tab:prognostic`)
- a joint feasibility table (`18-prognostic-joint-ceiling.R`)
- the statement that the stop-on-non-PD default came from "a prognostic scenario" with correlation 0.6 (lines 612-618)

Paper 15 must cite 01 for the DGP-level facts: the bound exists, and repair is refused. It should present only the analysis-level consequences. Paper 01 is at grade B and should not be reopened for this.

5.3 Portfolio count

The split adds a fifteenth manuscript. `docs/66` recommends consolidating toward about nine. The split is justified because the boundary is scientific, not editorial. It should be recorded alongside the merges already proposed there: 13 with 14, 08 into 10, 12 into 01, and 11 into 01 or 03.

6. Order of work

1. Decide the open question in Section 4 (the F-family member).
2. Add `--scope` to the ladder script and regenerate the predictive-only tables and figures.
3. Cut and rewrite paper 02 per 2.2. Re-verify every inline number. Run `perl tools/notation-lint.pl`. Render through `tools/render.sh`.
4. Decide 2.6 (a) or (b). If (b), run the large-N null.
5. Create `analysis/report/15-prognostic-biomarker/` and move the excised Rmd sections into a first draft, keeping their chunks working against the same data.
6. Implement and run the bm:t block (3.1).
7. Update CLAUDE.md, NOTATION.md, `docs/60` and `docs/66`.

7. Unverified assumptions in this plan

- The line ranges come from a grep of section headers and F1 and prognostic mentions, not a line-by-line read. Some cross-references outside the listed ranges will also need cutting.
- I have not checked whether `m2-pooled-size.csv` excludes the $c_{pb} = 0.3$ null cells (2.3).
- The large-N run cost in 2.6(b) is a rough estimate, not a timing.
- The cost of the bm:t block is scaled from block T's runtime. Its findings rest on a 300-replicate G8-only local pilot.